

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2018

Course: S.Y.B. Tech (All Branches)

Semester: III

Subject Name: Engineering Mathematics-III

Subject Code: BTBSC301

Max Marks:60

Date:30/11/2018

Duration: 03 Hrs

Instructions to the Students:

1. Attempt Any Five questions of the following. All questions carry equal marks.
2. Use of non-programmable scientific calculators is allowed.
3. Figures to the right indicate full Marks.

Q. 1. a) Show that,

$$\int_0^{\infty} \frac{\sin at}{t} dt = \frac{\pi}{2}$$

[4]

b) Find the Laplace transform of

$$\int_0^t \frac{e^{-3u} \sin 2u}{u} du$$

[4]

c) Find the Laplace transform of the function

$$f(t) = \begin{cases} 2 & , 0 < t < \pi \\ 0 & , \pi < t < 2\pi \\ \sin t & , t > 2\pi \end{cases}$$

[4]

Q.2. a) Find the inverse Laplace transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$.

[4]

b) By convolution theorem, find inverse Laplace transform of

$$\frac{s}{(s^2 + 1)(s^2 + 4)}$$

[4]

c) By Laplace transform method, solve the following simultaneous equations

$$\frac{dx}{dt} - y = e^t, \quad \frac{dy}{dt} + x = \sin t; \text{ given that } x(0) = 1, y(0) = 0.$$

[4]

Q. 3. a) Find the Fourier transform of

$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0 & , |x| > 1. \end{cases}$$

[4]

b) Find the Fourier sine transform of $e^{-|x|}$, and hence show that

$$\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}, \quad m > 0.$$

[4]

c) Using Parseval's Identity, prove that

$$\int_0^{\infty} \frac{t^2}{(t^2 + 1)^2} dt = \frac{\pi}{4}$$

[4]

Q.4. a) Solve the partial differential equation

$$(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$$

[4]

b) Use method of separation of variables to solve the equation

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u; \text{ given that } u(x, 0) = 6e^{-3x}$$

[4]

c) Find the temperature in bar of length 2 units whose ends are kept at zero temperature and lateral surface insulated if initial temperature is

$$\sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right)$$

[4]

Q. 5. a) If $f(z)$ is analytic function with constant modulus, show that $f(z)$ is constant.

[4]

b) If the stream function of an electrostatic field is $\psi = 3xy^2 - x^3$, find the potential function ϕ , where $f(z) = \phi + i\psi$.

[4]

c) Prove that the inversion transformation maps a circle in the z -plane into a circle in w -plane or to a straight line if the circle in the z -plane passes through the origin.

[4]

Q.6. a) Evaluate $\oint_C \frac{e^z}{(z-2)} dz$, where C is the circle $|z| = 3$.

[4]

b) Evaluate $\oint_C \tan z dz$, where C is the circle $|z| = 2$.

[4]

c) Evaluate, using Cauchy's integral formula:

[4]

$$1) \oint_C \frac{\cos(\pi z)}{(z^2 - 1)} dz \text{ around a rectangle with vertices } 2 \pm i, -2 \pm i.$$

$$2) \oint_C \frac{\sin^2 z}{(z - \frac{\pi}{6})} dz, \text{ where } C \text{ is the circle } |z| = 1.$$

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE End Semester Examination – Winter 2019 Course: B. Tech in Sem: III Subject Name: Engineering Mathematics-III (BTBSC301) Marks: 60 Date: 10/12/2019 Duration: 3 Hr.			
Instructions to the Students: 1. Solve ANY FIVE questions out of the following. 2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Attempt the following.		12
A)	Find $L\left\{\cos ht \int_0^t e^u \cosh u \, du\right\}$.	Analysis	4
B)	If $f(t) = \begin{cases} t, & 0 < t < \pi \\ \pi - t, & \pi < t < 2\pi \end{cases}$ is a periodic function with period 2π . Find $L\{f(t)\}$.	Analysis	4
C)	Using Laplace transform evaluate $\int_0^\infty e^{-at} \frac{\sin^2 t}{t} \, dt$	Evaluation	4
Q. 2	Attempt any three of the following.		12
A)	Using convolution theorem find $L^{-1}\left\{\frac{1}{s(s+1)(s+2)}\right\}$	Application	4
B)	Find $L^{-1}\{\bar{f}(s)\}$, where $\bar{f}(s) = \log\left(\frac{s^2+1}{s(s+1)}\right)$	Analysis	4
C)	Using Laplace transform solve $y'' + 2y' + 5y = e^{-t} \sin t$; $y(0) = 0$, $y'(0) = 1$	Application	4
D)	Find $L^{-1}\left\{\frac{s^2+2s-4}{(s-5)(s^2+9)}\right\}$	Analysis	4
Q. 3	Attempt any three of the following.		12

A)	Express the function $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ 0, & x > \pi \end{cases}$ as a Fourier sine integral and hence evaluate that $\int_0^\infty \frac{\sin \lambda x \sin \lambda \pi}{1-\lambda^2} d\lambda$.	Evaluation	4
B)	Using Parseval's identity for cosine transform, evaluate $\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)}$.	Application	4
C)	Find the Fourier sine transform of $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & x > 2 \end{cases}$.	Analysis	4
D)	If $F_s\{f(x)\} = \frac{e^{-as}}{s}$, then find $f(x)$. Hence obtain the inverse Fourier sine transform of $\frac{1}{s}$.	Analysis	4
Q. 4	Attempt any three of the following.		12
A)	Form the partial differential equation by eliminating arbitrary function f from $f(x^2 + y^2 + z^2, 3x + 5y + 7z) = 0$	Synthesis	4
B)	Solve $pz - qz = z^2 + (x + y)^2$	Application	4
C)	Determine the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ where the boundary conditions are $u(0, t) = 0$, $u(l, t) = 0$ ($t > 0$) and the initial condition $u(x, 0) = x$; l being the length of the bar.	Analysis	4
D)	Use the method of separation of variables to solve the equation $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, given that $u(x, 0) = 6e^{-3x}$	Application	4
Q. 5	Attempt the following.		12
A)	Determine the analytic function $f(z)$ in terms of z whose real part is $\frac{\sin 2x}{\cosh 2y - \cos 2x}$	Analysis	4
B)	Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. Find a function v such that $f(z) = u + iv$ is analytic.	Analysis	4
C)	Find the bilinear transformation which maps the points $z = 0, -1, -i$ onto the points $w = i, 0, \infty$. Also, find the image of the unit circle $ z = 1$.	Analysis	4
Q. 6	Attempt the following.		12

A)	Use Cauchy's integral formula to evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $ z = 3$.	Evaluation	4
B)	Find the poles of function $\frac{z^2 - 2z}{(z+1)^2(z^2+4)}$. Also find the residue at each pole.	Analysis	4
C)	Evaluate $\oint_C \frac{e^z}{\cos \pi z} dz$, where C is the unit circle $ z = 1$.	Evaluation	4
*** Paper End ***			

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – May 2019

Course: B. Tech

Sem: III

Subject Name: Engineering Mathematics-III

Subject Code: BTBSC301

Max Marks: 60

Date: 28-05-2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Attempt any three.		12
A) Find $L\{f(t)\}$, where $f(t) = t^2 e^{-3t} \sinh at$	Understand	4
B) Express $f(t)$ in terms of Heaviside's unit step function and hence find its Laplace transform where $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ \sin t, & t > \pi \end{cases}$	Understand	4
C) Find $L\{f(t)\}$, where $f(t) = 2^t \int_0^t \frac{\sin 3u}{u} du$	Understand	4
D) By using Laplace transform evaluate $\int_0^\infty e^{-t} \left(\frac{1 - \cos 2t}{t} \right) dt$	Evaluation	4
Q. 2 Attempt the following.		12
A) Using convolution theorem find $L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$	Application	4
B) Find $L^{-1} \{ \bar{f}(s) \}$, where $\bar{f}(s) = \cot^{-1} \left(\frac{s+3}{2} \right)$	Application	4
C) Using Laplace transform solve $y'' - 3y' + 2y = 12e^{-2t}$; $y(0) = 2$, $y'(0) = 6$	Application	4
Q. 3 Attempt any three.		12
A) Express $f(t) = \begin{cases} 1, & 0 \leq x \leq \pi \\ 0, & x > \pi \end{cases}$ as a Fourier sine integral and hence deduce that $\int_0^\infty \frac{1 - \cos \pi \lambda}{\lambda} \sin \pi \lambda d\lambda = \frac{\pi}{4}$.	Evaluation	4
B) Using Parseval's identity for cosine transform, prove that $\int_0^\infty \frac{\sin at}{t(a^2+t^2)} dt = \frac{\pi}{2} \left(\frac{1 - e^{-a^2}}{a^2} \right)$	Application	4

- C) Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$. Hence prove **Understand** **4**
that $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx = -\frac{3\pi}{16}$

- D) Find Fourier sine transform of $5e^{-2x} + 2e^{-5x}$ **Understand** **4**

Q. 4 Attempt the following.

- A) Form the partial differential equation by eliminating arbitrary function f from $f(x + y + z, x^2 + y^2 + z^2) = 0$ **Synthesis** **4**

- B) Solve $xz(z^2 + xy)p - yz(z^2 + xy)q = x^4$ **Analysis** **4**

- C) Find the temperature in a bar of length two units whose ends are kept at zero temperature and lateral surface insulated if the initial temperature is **Application** **4**
 $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.

Q. 5 Attempt Any three. **12**

- A) If the function $f(z) = (x^2 + axy + by^2) + i(cx^2 + dxy + y^2)$ is analytic, **Understand** **4**
find the values of the constants a, b, c and d .

- B) If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is **Understand** **4**
constant.

- C) Find the bilinear transformation which maps the points $z = 0, -i, -1$ into **Understand** **4**
the points $w = i, 1, 0$.

- D) Prove that the function $u = e^x(x \cos y - y \sin y)$ satisfies the Laplace's **Synthesis** **4**
equation. Also find the corresponding analytic function.

Q. 6 Attempt ANY TWO of the following. **12**

- A) Evaluate $\oint_C \frac{z+4}{z^2+2z+5} dz$, where C is the circle $|z+1-i|=2$. **Evaluation** **6**

- B) Find the residues of $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $|z|=2$. **Understand** **6**

- C) Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z|=3$. **Evaluation** **6**

*** End ***

Winter Examination – 2022

Course: - B. Tech.

Branch: - Common for All branches

Semester:- III

Subject Code & Name: BTBS301

Engineering Mathematics-III

Max. Marks: - 60

Date: - 09/03/2023

Duration: - 3-Hrs

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Three of the following.		12
A) Find Laplace Transform of $e^{-3t} \sin^2 t$	L3/CO1	4
B) Find Laplace Transform of $f(t) = \begin{cases} 1 & 0 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$ where $f(t)$ is periodic function of period 2.	L3/CO1	4
C) Evaluate using Laplace Transform.: $\int_0^\infty \frac{\cos 4t - \cos 3t}{t} dt$	L3/CO1	4
D) Find Laplace Transform of $(1 + 2t - 3t^2 + 4t^3)H(t - 2)$	L3/CO1	4
Q2 Solve Any Three of the following.		12
A) Find the inverse Laplace transformation of the function. $\log \left(1 + \frac{a^2}{s^2} \right)$	L3/CO2	4
B) By using convolution theorem find $L^{-1} \left[\frac{s}{(s^2+4)(s^2+9)} \right]$	L3/CO2	4
C) Find the inverse Laplace transformation of the function. $\frac{5s^2-15s-11}{(s+1)(s-2)^2}$	L3/CO2	4
D) Solve using Laplace transformation $y'' + 3y' + 2y = t\delta(t - 1)$ for which $y(0) = y'(0) = 0$	L3/CO2	4

Q.3 Solve Any Three of the following.**(12)**

A) Using Parseval's identity prove that $\int_0^\infty \frac{x^2}{(x^2+1)^2} dx = \frac{\pi}{4}$ L3/CO3 **4**

B) Find the Fourier transform of

$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0 & , \quad |x| > 1 \end{cases} \quad \text{L3/CO3} \quad \mathbf{4}$$

C) Find the Fourier Sine transform e^{-ax} , $a > 0$ L3/CO3 **4**

D) Find the Fourier cosine transform of the function $f(y) = \begin{cases} \cos y, & 0 < y < a \\ 0, & y > a \end{cases}$ L3/CO3 **4**

Q.4 Solve Any Three of the following.**(12)**

A) Form the partial differential equation by eliminating arbitrary constants from L3/CO4 **4**

$$(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$$

B) Solve the Partial differential equation $x(y - z)p + y(z - x)q = z(x - y)$ L3/CO4 **4**

C) Use the method of separation of variables to solve

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u \quad \text{given that } u(x, 0) = 6e^{-3x} \quad \text{L3/CO4} \quad \mathbf{4}$$

D) A bar with insulated at its ends is initially at temperature 0°C throughout. The end $x = 0$ is kept at 0°C for all times and the heat is suddenly applied so that $\frac{\partial u}{\partial x} = 10$ at $x = t$ for all time. Find the temperature function $u(x, t)$ L3/CO4 **4**

Q.5 Solve Any Three of the following.**(12)**

A) Determine k such that the function $f(z) = e^x \cos y + ie^x \sin ky$ is analytic. L3/CO5 **4**

B) Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and L3/CO5 **4**
hence determine the analytic function $f(z)$ in terms of z .

C) Determine the pole of the function $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ and also find the residue at each pole

& sum of all residues. L3/CO5 **4**

D) Evaluate L3/CO5 **4**

$$\oint_C \frac{\sin \pi z^2 + 2z}{(z-1)^2(z-2)} dz, \text{ Where } C \text{ is the circle } |z| = 4$$

*** End ***

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Semester: III Subject Code & Name: BTBS301(Engineering Mathematics III) Max Marks: 60 Date: Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Find the Laplace Transform of $e^{4t} \sin^3 t$	CO1	6
B)	Evaluate $\int_0^\infty \frac{\cos at - \cos bt}{t} dt$ by using Laplace transform.	CO1	6
C)	Express in terms of Heaviside unit step function and find its Laplace transform. $f(t) = \sin t$, for, $0 < t < \pi$ $= \sin 2t$, for, $\pi < t < 2\pi$ $= \sin 3t$, for, $t > 2\pi$	CO1	6
Q.2	Solve Any Two of the following.		
A)	Find the inverse Laplace transform of $\log\left(\frac{s+a}{s+b}\right)$	CO2	6
B)	Find the inverse Laplace transform of $\frac{5s^2-15s-11}{(s+1)((s-2)^2)}$	CO2	6
C)	Solve using Laplace transform $3 \frac{dy}{dt} + 2y = e^{3t}, y = 1$ at $t = 0$.	CO2	6
Q. 3	Solve Any Two of the following.		
A)	Find the Fourier Transform of $f(x) = 1, \text{for } x < 1$ $= 0, \text{for } x > 1$ Hence evaluate that $\int_0^\infty \frac{\sin x}{x} dx$.	CO3	6
B)	Find the Fourier cosine transform of e^{-x^2}	CO3	6
C)	Using Parseval's Identity, prove that $\int_0^\infty \frac{t^2}{(t^2+1)^2} dt = \frac{\pi}{4}$	CO3	6
Q.4	Solve Any Two of the following.		

A)	Form the partial differential equation by eliminating the arbitrary function from: $f(x + y + z, x^2 + y^2 + z^2)$	CO4	6
B)	Solve the partial differential equation: $(mz - ny)p + (nx - lz)q = ly - mx$.	CO4	6
C)	If the initial displacement and velocity of a string stretched between $x = 0$ & $x = l$ are given by $y = f(x)$ & $\frac{dy}{dt} = g(x)$, determine the displacement y of any point at a distance x from one end at time t .	CO4	6
Q. 5	Solve Any Two of the following.		
A)	If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .	CO5	6
B)	Find the bilinear transformation that maps the points $z=0, -1, i$ into the points $w=i, 0, \infty$ respectively.	CO5	6
C)	Use Cauchy's Integral formula to evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, Where C is the circle $ z = 2$	CO5	6
	*** End ***		

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Supplementary Summer 2023 Course: B. Tech. Branch :All Branches Semester :III Subject Code & Name: BTES301 Engineering Mathematics-III Max Marks: 60 Date:08/08/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Three of the following.		12
A)	Find $L\{e^{-t}t\sin 2t\cos 2t\}$	Understand	4
B)	Find $L\left\{e^{-2t}\int_0^t \frac{\sin 3t}{t} dt\right\}$	Understand	4
C)	Evaluate using Laplace transform $\int_0^\infty e^{-t} \left[\frac{\cos 5t - \cos t}{t}\right] dt$	Application	4
D)	Express the following function in Heaviside's unit step function and find it's Laplace transform $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \\ 0, & t > 3 \end{cases}$	Application	4
Q.2	Solve Any Three of the following.		12
A)	Find $L^{-1}\left[\frac{s-1}{s(s+2)(s-3)}\right]$	Application	4
B)	Find $L^{-1}\left[\frac{1}{(s+4)(s^2+1)}\right]$ by convolution theorem	Application	4
C)	Find $L^{-1}\left\{\log\left[\frac{s^2-4}{(s-2)^2}\right]^{1/5}\right\}$	Application	4
D)	Solve $y'' - 6y' + 9y = t^2 e^{3t}$, where $y(0) = 2, y'(0) = 6$	Application	4
Q. 3	Solve Any Three of the following.		12
A)	Find the Fourier integral representation of the function $f(x) = \begin{cases} e^{ax}, & x \leq 0 \\ e^{-ax}, & x \geq 0 \end{cases}$ and show that $\int_0^\infty \frac{\cos \lambda x}{a^2 + \lambda^2} d\lambda = \frac{\pi}{2a} e^{-ax}$	Evaluation	4
B)	Find the Fourier cosine transform of $f(x) = e^{-x} + e^{-2x}, x > 0$	Evaluation	4
C)	Find the Fourier sine transform of $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & x > 2 \end{cases}$		4

D)	Solve the integral equation $\int_0^{\infty} f(x) \cos \omega x \, dx = \begin{cases} 1 - \omega, & 0 < \omega < 1 \\ 0, & \omega > 1 \end{cases}$	Evaluation	4
Q.4	Solve Any Three of the following.		12
A)	Form the partial differential equation by eliminating the arbitrary function from $(x - h)^2 + (y - k)^2 + z^2 = a^2$, h & k being constants	Understand	4
B)	Solve the partial differential equation $\frac{y - z}{yz} p + \frac{z - x}{zx} q = \frac{x - y}{xy}$	Application	4
C)	Using the method of separation of variables, solve $\frac{\partial u}{\partial x} + u = \frac{\partial u}{\partial t}$, if $u = 4e^{-3x}$, when $t = 0$	Application	4
D)	Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ under the conditions $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = 0$, $u(l, t) = 0$, where $0 < x < l$.	Application	4
Q. 5	Solve Any Three of the following.		12
A)	Prove that $v = e^x \cos y + x^3 - 3xy^2$ is harmonic function. Find its harmonic conjugate and corresponding analytic function.	Understand	4
B)	Find the analytic function whose imaginary part is $r^2 \cos 2\theta - r \cos \theta + 2$	Understand	4
C)	Evaluate $\int_c \frac{z}{(z - 1)(z - 2)^2} dz$, if c is the circle $ z - 2 = \frac{3}{2}$ by using Cauchy residue theorem.	Evaluation	4
D)	Evaluate $\int_c \frac{3z + 4}{z(2z + 1)} dz$, if c is the circle $ z = 1$ by using Cauchy Integral theorem.	Evaluation	4
*** End ***			

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	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2023 Course: B. Tech. (Common to all Branches) Semester : III Subject Name & Code: Engineering Mathematics – III (BTBSC 301) Max Marks: 60 Date:08/08/2023 Duration: 3 Hrs.	
	Instructions to the Students: 1. Attempt any FIVE of the following questions. 2. All questions carry equal marks. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		Marks
Q. 1	Solve Any Two of the following.	12
A)	Find the Laplace transform of $F(t) = \frac{e^{-at} - e^{-bt}}{t}$	6
B)	Find the Laplace transform of $F(t) = \sin 2t \cos 3t$	6
C)	Find the Laplace transform of $\operatorname{erf}(\sqrt{t})$.	6
Q.2	Solve Any Two of the following:	12
A)	State and prove the convolution theorem for finding the inverse Laplace transform.	6
B)	Using Partial Fraction method, find the inverse Laplace transform of $\bar{f}(s) = \frac{5s+3}{(s-1)(s^2+2s+5)}$	6
C)	Find the inverse Laplace transform of $\bar{f}(s) = \cot^{-1}\left(\frac{s+3}{2}\right)$	6
Q. 3	Solve any Two of the following:	12
A)	Find the Fourier sine transform of $e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$	6
B)	Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & x \leq 1 \\ 0, & x > 1 \end{cases}$. Hence evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$.	6
C)	Evaluate the integral $\int_0^\infty \frac{t^2}{(t^2+1)^2} dt = \frac{\pi}{4}$.	6
Q.4	Solve any Two of the following:	12
A)	The partial differential equation by eliminating the arbitrary function from $z = x + y + f(xy)$	6
B)	The partial differential equations by eliminating the arbitrary constant $z = (x^2 + a)(y^2 + b)$	6
C)	Solve the following partial differential equations $(mz - ny)p + (nx + lz)q = ly - mx$ where the symbols have got their usual meanings.	6
Q. 5	Solve any Two of the following:	12
A)	Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence determine the corre-	6

	sponding analytic function	
B)	If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant	6
C)	Under the transformation $W = \frac{1}{z}$, find the image of $ z - 2i = 2$.	6
Q. 6	Solve any Two of the following:	
A)	Evaluate $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x$ and $y = x^2$	6
B)	Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ where C is the circle $ z = 2$ and $ z = \frac{1}{2}$	6
C)	Use Cauchy's integral formula to evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C the circle is $ z = 2$.	6
	*** End ***	

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Branch : B. Tech (Common to all)

Semester : III

Subject with code: Engineering Mathematics – III (BTBS 301)

Max Marks: 60

Date: 08/08/2023

Duration: 3 Hr

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	Level/CO	Marks
Q. 1 Solve Any Two of the following.		12
A) Find the Laplace transform of $f(t) = \frac{e^t - \cos t}{t}$	Understand/ (CO1)	6
B) Using Laplace transform prove That $\int_0^\infty t e^{-3t} \sin t dt = \frac{3}{50}$	Understand/ (CO1)	6
C) Find the Laplace transform of the triangular wave function of period $2c$ given by $f(t) = \begin{cases} t, & 0 \leq t \leq c \\ 2c - t, & c < t < 2c \end{cases}$	Remember/ (CO1)	6
Q.2 Solve Any Two of the following.		12
A) Find the inverse Laplace transforms of $\bar{f}(s) = \frac{s e^{-4s}}{s^2+9}$	Understand/ (CO2)	6
B) By convolution theorem, find the inverse Laplace Transforms of $\bar{f}(s) = \frac{1}{s(s^2-a^2)}$	Understand/ (CO2)	6
C) Solve the equation $\frac{d^3y}{dt^3} + 2\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 0$, where $y = 1, \frac{dy}{dt} = 2, \frac{d^2y}{dt^2} = 2$ at $t = 0$, by Laplace transform method.	Remember/ (CO2)	6
Q. 3 Solve Any Two of the following.		12
A) Using the Fourier integral representations, show that $\int_0^\infty \frac{\cos x\omega}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x} \quad (x \geq 0)$	Understand/ (CO3)	6
B) Find the Fourier sine transform of $\frac{e^{-ax}}{x}$.	Understand/ (CO3)	6
C) Using Parseval's identity Evaluate $\int_0^\infty \frac{\sin^2 x}{x^2} dx$	Remember/ (CO3)	6

Q.4 Solve Any Two of the following.		12
A) Form the partial differential equation by eliminating the arbitrary functions from $z = f(x + it) + g(x - it)$	Understand/ (CO4)	6
B) Solve the partial differential equation $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$	Understand/ (CO4)	6
C) Use the method of separation of variables to solve the equation $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0.$	Remember/ (CO4)	6
Q.5 Solve Any Two of the following.		12
A) Find a function $w = u + iv$ which is analytic if $u = x^2 - y^2$.	Understand/ (CO5)	6
B) Evaluate $\int_C \frac{\cos \pi z^2}{(z-1)(z-2)} dz$, where C is $z = \frac{3}{2}$.	Understand/ (CO5)	6
C) By Residue theorem evaluate $\int_C \frac{dz}{(z^2+4)^2}$, where C is the circle $z - i = 2$.	Understand/ (CO5)	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
Regular & Supplementary Winter Examination-2023			
Course: B. Tech.	Branch: ALL	Semester: III	
Subject Code & Name: BTBS301/ BTES 301 Engineering Mathematics-III			
Max Marks: 60	Date: 02.01.2024	Duration: 3 Hr.	
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find the Laplace transform of $f(t) = t^2 \sin 2t$	Understand/ CO1	6
B)	Find Laplace transform of $F(t) = \int_0^t \frac{e^{-at} - e^{-bt}}{t} dt$	Understand /CO1	6
C)	Find the Laplace transforms of $f(t) = \frac{t}{T}$, for $0 < t < T$ (saw - tooth wave function of period T)	Apply/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Find inverse Laplace transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$	Understand /CO2	6
B)	By using Partial fraction expansion to find inverse Laplace transform of $F(s) = \frac{s}{(s^2+1)(s^2+4)}$	Understand /CO2	6
C)	Using the Laplace transform, solve the differential equation $\frac{d^2x}{dt^2} + 9x = \cos 2t$; if $x(0) = 1$, $x\left(\frac{\pi}{2}\right) = -1$.	Apply/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Express the function $f(x) = \begin{cases} 1 & \text{for } x \leq 1 \\ 0 & \text{for } x > 1 \end{cases}$ as a Fourier integral.	Understand /CO3	6
B)	Find the Fourier sine transform of $f(x) = e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$.	Understand /CO3	6
C)	Using Parseval's identity, show that $\int_0^\infty \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$	Apply/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Solve the following partial differential equations $(mx - ny)p + (nx - lz)q = ly - mx$	Understand /CO4	6

B)	A string is stretched and fastened to two points l apart. Motion is started by replacing the string in the form $y = A \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of a point at a distance x from one end at time t is given by $y(x, t) = A \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l}$.	Apply/CO4	6
C)	Solve the following equation by the method of separation of variables: $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$, given that $u = 0$ when $t = 0$ and $\frac{\partial u}{\partial t} = 0$ when $x = 0$.	Apply /CO4	6
Q. 5	Solve Any Two of the following.		12
A)	If $f(z)$ is analytic, show that $\left[\frac{\partial f(z) }{\partial x} \right]^2 + \left[\frac{\partial f(z) }{\partial y} \right]^2 = f'(z) ^2$.	Understand /CO5	6
B)	Apply Cauchy's integral Formula to evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle (a) $ z = 2$ and (b) $ z = \frac{1}{2}$.	Apply/CO5	6
C)	State Cauchy's residue theorem and evaluate $\oint_C \tan z dz$, where C is the circle $ z = 2$.	Apply /CO5	6
	*** End ***		